



Unit-1

Short Questions

- 1) Define Number System. List out different types of number systems?
- 2) Convert $(011101011)_2$ into an octal number.
- 3) What is 2's Complement? Give one example.
- 4) Define De Morgan's Law.
- 5) Convert $(A2E)_{16}$ to Octal number.
- 6) Convert $(76543)_8$ to Hexadecimal
- 7) Explain about NAND gate with symbol and truth table.
- 8) Explain about XNOR gate with symbol and truth table.
- 9) Express the Boolean function $F = A + B'C$ in a sum of minterms.
- 10) Express $F = xy + x'z$ in a product of maxterms form.

Long Questions

- 1) Convert $(243)_{10}$ into following number system:
a) Binary number b) Octal number c) Hexadecimal number.
- 2) Convert $(11101011)_2$ into following number system:
a) Octal number b) Decimal number c) Hexadecimal number.
- 3) Convert $(247)_8$ into following number system:
a) Binary number b) Decimal number c) Hexadecimal number.
- 4) Convert $(B2E)_{16}$ into following number system:
a) Binary number b) Octal number c) Decimal number
- 5) find the 2's complement for the following
a) $(01010101)_2$ b) $(10101101)_2$ c) $(0110110100)_2$
- 6) What is Boolean Algebra? Give any five properties of Boolean Algebra.
- 7) a) Express the Boolean function $F = (x + yz)$ as a sum of Minterms.
b) Express the Boolean function $F = (x + yz)$ as a product of Maxterms.
- 8) a) Express the Boolean function $F = (A+B'C+ABC)$ as a sum of Minterms.
b) Express the Boolean function $F = A'(B'+C)$ as a product of Maxterms.
- 9) What are Logic gates? Briefly explain about Basic logic gates with symbols and truth tables.
- 10) Explain briefly about Universal logic gates and Derived logic gates with symbols and truth tables.



Unit-2

Short Questions

- 1) What is the K-Map? what are the steps to solve expression using K-Map?
- 2) Simplify the Boolean function $F(A,B) = \sum m(0,1,2)$.
- 3) Simplify the Boolean function $F(A,B) = \pi M(0,1,3)$.
- 4) Simplify the Boolean function $F(A,B,C) = \sum m(0,1,2,3,5,7)$.
- 5) Simplify the Boolean function $F(A,B,C) = \pi M(0,2,3,4,5,6)$.
- 6) Simplify the following Boolean functions $F(A,B,C) = \sum m(0,1,2) + d(3,7)$.
- 7) Describe the implementation of XNOR gate using NAND gate.
- 8) Describe the implementation of XNOR gate using NOR gate.
- 9) Using Boolean algebra techniques, simplify the Boolean Expression

$$F = A'B'C + A'BC + AB'$$

- 10) Using Boolean algebra techniques, simplify this expression

$$F = A'.B.C + A.B'.C + A.B.C' + A.B.C$$

Long Questions

- 1) Simplify the Boolean functions using K-Map
 - a) $F(A,B) = \sum m(1,2)$.
 - b) $F(A,B) = \pi M(0,3)$.
- 2) Simplify the Boolean functions using K-Map
 - a) $F(A,B,C) = \sum m(1,2,3,7)$.
 - b) $F(A,B,C) = \pi M(0,1,3,5)$.
- 3) Simplify the Boolean functions using K-Map
 - a) $F(A,B,C) = \sum m(1,2,3,4,5,7)$.
 - b) $F(A,B,C) = \pi M(0,2,3,6,7)$.
- 4) Simplify the following Boolean Expressions
 - a) $F(A,B,C) = AB + (AC)' + AB'C(AB+C)$
 - b) $F(A,B,C) = (A+B)(A+C)$
- 5) Simplify the following Boolean Expressions
 - a) $F(A,B,C,D,E) = A + A'B + A'B'C + A'B'C'D + A'B'C'D'E$
 - b) $F(A,B,C) = (A+B)(A+C)$



- 6) Simplify the following Boolean functions
- a) $F(A,B,C) = \sum m(1,3,4,7) + d(2,5)$.
 - b) $F(A,B,C) = \pi M(0,1,3,6) + d(5,7)$.
- 7) Simplify the following Boolean functions
- a) $F(A,B,C) = \sum m(1,2,3,4,5) + d(6,7)$.
 - b) $F(A,B,C) = \pi M(0,3,5) + d(1,4,6)$.
- 8) Describe the implementation of the following gates using NAND gate
- a) AND gate b) OR gate c) NOT gate
- 9) Describe the implementation of the following gates using NOR gate
- a) AND gate b) OR gate c) NOT gate
- 10) Describe the implementation XOR Gate using a) NAND Gate b) NOR Gate

Unit-3

Short Questions

- 1) What is a Combinational logic circuit? Draw the Block diagram of combinational circuit.
- 2) Draw the Half adder circuit diagram.
- 3) Draw the Half Binary subtractor circuit diagram.
- 4) Write the differences between Binary adder and BCD adder.
- 5) Perform BCD addition of $(1000)_2$ and $(0111)_2$ and write the corrected BCD result.
- 6) Define a magnitude comparator. What are its three possible outputs?
- 7) What is 1-bit comparator and draw its truth table.
- 8) Write the difference between Encoder and Decoder.
- 9) Define a multiplexer (MUX). Why is it called a data selector?
- 10) Write the difference between multiplexer and demultiplexer.

Long Questions

- 1) Explain Full adder with logical diagram and truth table.
- 2) Explain Half Binary adder with logic diagram and truth table.
- 3) Explain Full Binary Subtractor with logic diagram and truth table.
- 4) Explain Half Binary Subtractor with logic diagram and truth table.
- 5) Explain the design and operation of a BCD Adder with a suitable block diagram.
- 6) Design and explain a 2-bit Comparator with its truth table and logic gates.



- 7) Design and explain 8x3 Encoder with its truth table and logic diagram.
- 8) Design and explain 3x8 Decoder with its truth table and logic diagram.
- 9) Explain the operation of a Multiplexer. Design and explain a 4-to-1 multiplexer using logic gates, including its truth table.
- 10) Explain the operation of a Demultiplexer. Design and explain a 1-to-4 demultiplexer with its truth table, logic diagram.

Unit-4

Short Questions

- 1) What is a latch?
- 2) What is a flip-flop?
- 3) What is a clocked S-R flip-flop?
- 4) What is a D flip-flop?
- 5) What is a T flip-flop?
- 6) What is a J-K flip-flop?
- 7) What is a master-slave J-K flip-flop?
- 8) What is a counter?
- 9) What is a ring counter?
- 10) What is a shift register?

Long Questions

- 1) Explain the working of an S-R flip-flop with a neat diagram and truth table?
- 2) Explain the working of a D flip-flop with a neat diagram and truth table?
- 3) Explain the working of a T flip-flop with a neat diagram and truth table?
- 4) Explain the working of a J-K flip-flop with a neat diagram and truth table?
- 5) Explain the working of a Master-Slave J-K flip-flop with a neat diagram?
- 6) Write the differences between synchronous and asynchronous counters.
- 7) Explain the working of a 3-bit asynchronous counter with a neat diagram and counting sequence?
- 8) Explain the working of a ring counter with a neat diagram and counting sequence?
- 9) Explain Left-shift register with suitable diagrams with example?
- 10) Explain Right-shift register with suitable diagrams with example?



Unit-5

Short Questions

- 1) What is a memory device?
- 2) What is RAM?
- 3) What is ROM?
- 4) What is PROM?
- 5) What is EPROM?
- 6) What is EEPROM?
- 7) What is EAPROM?
- 8) What is PLA?
- 9) What is PAL?
- 10) What is the difference between RAM and ROM?

Long Questions

- 1) Explain RAM and its basic working with a neat diagram?
- 2) Explain ROM and its basic working with a neat diagram?
- 3) Explain PROM and how it is programmed?
- 4) Explain EPROM and its working in simple steps?
- 5) Explain EEPROM and its working in simple steps?
- 6) Explain EAPROM and mention its main features?
- 7) Write short notes on ROM, PROM, EPROM, and EEPROM?
- 8) Explain the basic structure and working of a Programmable Logic Array (PLA)?
- 9) Explain the differences between PLA and PAL with a simple comparison table?
- 10) Explain the basic structure and working of Programmable Array Logic (PAL)?